Roberto Arturo González

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Quantum Hardware Technician | Vacuum Systems, Optics, Photonics, Precision Assembly

Demonstrating systems troubleshooting, precise assembly, complete vacuum system rebuilds, optical table alignment and helium leak detection ensuring high efficient performance operation across complex technical environments. Known for dependability, adaptability and working effectively in technical & non/technical team environments.

Technical & Essential Skills

- Mechanical & Systems: Preventative maintenance, troubleshooting on refrigeration systems
- Fabrication/Tools: MIG welding, Soldering certified, hand/power tools, milling, 3D printing
- Engineering Skills: SolidWorks (design/modeling)
- Quantum Skills: Ultra High vacuum pump systems, helium leak checking, photonics, optics, laser safety, photon detectors, proper mirror/lens cleaning procedures, alignment, optomechanics, basic electronics
- Operations: Data entry, inventory control, reporting, logistics, documentation
- Instrumentation: Oscilloscopes, power meters, spectrometers, voltmeters

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Project Experience

Thorlabs Optical Tweezers Kit

- Using a 639 nm focused beam to trap, manipulate or move a microscopic silica bead
- Worked in a group to build, troubleshoot and demonstrate to visitors & middle school students

Edwards Vacuum Pump

- Complete disassembly and reassembly of vacuum pump system as well as rotary vane pump
- Baseline testing conducted to validate proper function after tear down/reassembly
- Helium leak detection system used to ensure all fittings and valves are functional, torqued and sealed properly

Thorlabs Quantum Optics Kit

- Complete assembly of Optics kit using Thorlabs laser, optomechanics, optics, single photon detectors & BBO crystal
- Mounting various lenses or mirrors and ensuring proper direction, height and alignment for the most accurate readings on any power meter
- Calculating count rates and frequencies, we used time tagging electronics to a PC with the Thorlabs EDU-QOP1 software to document data

Saturated Absorption Spectroscopy

- Built, troubleshooted saturated absorption spectroscopy of Rubidium vapor
- Gathered data on oscilloscope

Technical Training

Quantum Technician Hardware Skills

- Fundamentals of quantum science, physics & technology
- Full time Monday-Friday 8-5 intensive project based training program led by Dr. Brian Rashap
- Over 400hrs of hands on technical training & lecture, planning/completing complex development projects as a group
- High Vacuum & Ultra High Vacuum system operation, disassembly, assembly, base line testing & trouble shooting, leak checking bakeout on Turbo, Ion, and Scroll pumps
- Optical system assembly & alignment: lasers, lenses, waveplates, beamsplitters and single photon detectors, photo detectors, acousto-optical modulators (ADM's)
- Fiber coupling, alignment, cleaning practices on fibers & mirrors
- Technical documentation, mathematics, statistics and technical presentations

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Education

Deep Dive Quantum Technician Bootcamp, Central New Mexico Community College

Albuquerque High School, Class of 2015, State of New Mexico & Albuquerque Public Schools Bilingual Seal Recipient

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Work Experience

Ben E Keith Co. **Refrigeration Specialist/Dematic Technician** **2021- 2026**

- Managed Anhydrous Ammonia system for entire warehouse facility
- Performed routine preventative/reactive maintenance on both Dematic & Ammonia Refrigeration system including replacement/rebuilding valves, complete valve station adjustments, solenoids, various electrical components on Dematic Multiflex Shuttles, precise laser adjustment, homing calibration, teaching calibration
- Installation of motorized fingers, motors, sensors and communications components
- Welded & fabricated multiple structural components inside machine room of Ammonia system & warehouse facility
- Hazwopper and chemical training, safety data sheet practice, proper protective equipment knowledgeable, industrial electric forklift certified
- Mentor/Trained 1-2 interns

Kelly Cable of New Mexico **Field Surveyor** **2022- 2023**

- Captured GPS related photographs of telecommunication poles for measurement & location purposes
- Created powerpoint presentation packages of documented locations & measurements
- Easily cooperated with drafting engineers to complete customers projects with proper communication in timely manners

Shamrock Foods Co **Truck loader/Order Selector** **2017-2021**

- Guaranteed non-cross contamination within loads before delivery
- Independently ensured proper temperature & food safety departures

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